Markov Chains in Finance

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Abstract

This paper details the method of finding the long-term probability distribution of a financial market being in a period of increasing, decreasing or stable stock prices. For this, a transition matrix will be derived from weekly data, and then the stationary distribution will be determined, finally, it will be proven to be the long-term probability distribution of the Markov Chain in question. The average time to transition to an increasing market will also be derived.

1 Introduction

The purpose of this project is to investigate the behavior of the S&P 500 index over the period from January 5, 2013, to December 30 2022, focusing on the weekly returns. The objective is to classify the market into distinct states — namely bull, bear, and stable markets — based on the weekly percentage change in the index. By examining these states, we aim to understand the transition probabilities between different market conditions and how the market behaves in each state.

2 Preliminary definitions

Definition 2.1 (Discrete Markov Chain)**.** A random process $(X_n)_{n=1}^{\infty}$ with countable state space S is called a Markov Chain if

$$\mathbb{P}(X_{n+1} = k \mid X_n = j, X_{n-1} = i_{n-1}, ..., X_0 = i_0) = \mathbb{P}(X_{n+1} = k \mid X_n = j)$$

holds for any $n \geq 0$ and $k, j, i_{n-1}, ..., i_0 \in S$.

Definition 2.2 (Transition Matrix)**.** The array $\mathbf{P} = (p_{ij})_{i,j \in S}$ is called a transition matrix for the Markov Chain where $\sum_{j \in S} p_{ij} = 1$ and $p_{ij} \geq 0$ for all $i, j \in S$. ($\mathbf{P}$ is a stochastic matrix.)

Definition 2.3 (Irreducibility)**.** A Markov Chain is irreducible if the state space S is irreducible, i.e. all states in S are recurrent, and they intercommunicate.

Definition 2.4 (Period of a state)**.** Let j be a recurrent state. The period $d(j)$ of j is the greatest common divisor of the times at which return to j is possible, i.e.

$$d(j) = \gcd(n \geq 1 : p_{jj} > 0)$$

Definition 2.5 (Aperiodicity)**.** A Markov Chain is aperiodic if all states are aperiodic, i.e. $d(j) = 1$ for all $j \in S$.

Definition 2.6 (Financial Market)**.** A financial market is a market in which one or more financial assets are traded. They include *primary markets* (where new securities are issued), *secondary markets* (where previously issued securities are traded), and *over-the-counter markets* (that trade issued securities that are not traded on an exchange).

Definition 2.7 (Bull Market)**.** A bull market refers to a period in which stock prices rise for an extended time. In this case when prices are up by at least 1%.

Definition 2.8 (Bear Market)**.** A bear market refers to a period in which stock prices fall for an extended time. In this case when prices are down by at least 1%.

Definition 2.9 (Stable Market)**.** A stable market refers to a period in which stock prices remain stable for an extended time. In this case when prices vary between $\pm 1\%$.

3 Data collection

The dataset consists of the weekly closing price (Note:Unadjusted) of the S&P 500 index, specifically taken from the last trading day of each week (Friday), from 4 January 2013, to 30 December 2022 (totaling 521 weeks). The data was collected using the Yahoo finance library.

Let the state space $S = \{$'bull', 'bear', 'stable'$\}$. To determine the transition matrix $\mathbf{P}$, we process the data accordingly (details can be found in the appendix, code section). The data was categorized based on the percentage change in price:

Change in the price in %	Assigned category
< -1	Bear $(-)$
-1 to 1	Steady (0)
> 1	Bull $(+)$

3.1 Weekly Return

To determine whether the market was in a bull, bear, or steady state on a weekly basis, we calculated the weekly return using the formula

$$\text{Return} = \frac{\text{Close}_{\text{this week}} - \text{Close}_{\text{last week}}}{\text{Close}_{\text{last week}}} \times 100.$$

The data was then classified into the three categories based on the weekly returns, as outlined above in state space $\mathbf{S}$.

3.2 Transition Matrix Calculation

The transition probabilities from state i to state j were derived using the following method: Let n_i be the number of transitions from i and let m_{ij} be the number of transitions from i to j. Then

$p_{ij} = \frac{m}{n}$. (Note that the last data point will not be counted towards n, since it does not transition into another state.)

Example Let the data points be

$$++0-0--+0+-++00+-+-0.$$

Then $n_- = 6$, $n_0 = 5$ (since the last 0 is not counted), and $n_+ = 8$ Then the probabilities for the first row are calculated as follows:

bear to bear	$m_{--} = 1$	$p_{--} = 1/6$
bear to stable	$m_{-0} = 2$	$p_{-0} = 2/6$
bear to bull	$m_{-+} = 3$	$p_{-+} = 3/6$

Using this method, we constructed a 3×3 transition matrix. The transition matrix for the S&P 500 market over the period from January 4, 2013, to December 30, 2022, is as follows:

$$\mathbf{P} = \left(\begin{array}{c|ccc} & - & 0 & + \\ \hline - & 0.230769 & 0.317308 & 0.451923 \\ 0 & 0.190871 & 0.522822 & 0.286307 \\ + & 0.194286 & 0.468571 & 0.337143 \end{array} \right)$$

- Number of Bullish weeks: $175 \to 175/521 \approx 33.59\%$ of the time, the market was bull state.
- Number of Bearish weeks: $104 \to 104/521 \approx 19.96\%$ of the time, the market was bear state.
- Number of Steady weeks: $242 \to 242/521 \approx 46.45\%$ of the time, the market was stable state.

The market was stable most of the time, given the $\pm 1\%$ threshold. The least amount of time was spent in a bear state, indicating that the market was relatively positive over the 10-year period analyzed. Figure 1 shows the time series plot of weekly returns, illustrating the movement of the S&P 500 index on a weekly basis over the 10-year period. Notably, the market exhibited significant volatility between 2020 and 2022, which may influence subsequent calculations.

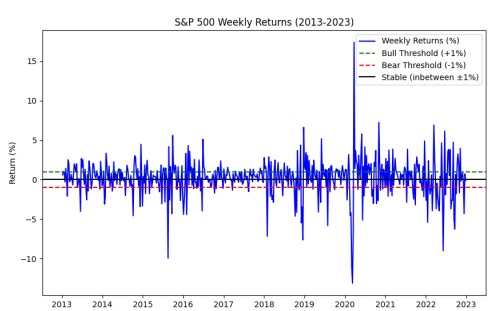

Figure 1: Time series plot of weekly returns.

4 Stationary Distribution and Long-Term Behavior

4.1 Transition Matrix P after n Steps

Given that the market is in a bearish state this week, we aim to determine the probability that the market will be in a bullish, bearish, or stable state after n weeks. This can be achieved using the Chapman-Kolmogorov equations, which allow us to compute the probability distribution of future states based on the current state and the transition matrix.
By raising the transition matrix $\mathbf{P}$ to the power of n, we can determine the probabilities of the market transitioning from its current state to any other state after n weeks. Specifically, the row of $\mathbf{P}^n$ corresponding to the current state provides the probabilities of transitioning to each possible future state. This approach enables us to model the expected market behavior over various time horizons.

4.2 Transition Matrix in the Long-Run

To determine the long-run probability distribution of the market states, we examine the stationary distribution of the Markov chain. The Markov chain under consideration is irreducible, as it is possible to transition from any state to any other state in a finite number of steps. Additionally, the chain is aperiodic, and the state space is finite. These properties guarantee the existence of a unique stationary distribution π satisfying the balance equation:

$$\pi \mathbf{P} = \pi,$$

where the elements of π sum to one, $\sum \pi_i = 1$. Solving for π, we obtain:

$$\pi = (0.2, 0.4635, 0.3365).$$

This implies that, in the long run, the market has a 20% of being in a bearish state,a 46.35% probability of being in a stable state, and a 33.65% probability of being in a bullish state. The calculations demonstrate the long-run stability of the market, reinforcing the observation that the market is generally more inclined toward a stable state rather than prolonged bearish phases.

4.3 Expected Time to Reach a Bullish State

To determine the average time required for a bearish or stable market to transition to a bullish state, we employ the concept of mean first passage times in Markov chains. The mean first passage time m_{jk} is the expected number of steps it takes to reach state k for the first time, starting from state j.

Definition 4.1 (Hitting time)**.** The first passage time from state j to state k for a Markov chain is

$$T_{jk} = \min\left(n \geq 1 : X_n = k \mid X_0 = j\right).$$

Recall:

- j is recurrent if

$$P(T_{jj} < \infty) = \sum_{n=1}^{\infty} P(T_{jj} = n) = 1.$$

- j is transient if

$$\sum_{n=1}^{\infty} P(T_{jj} = n) < 1.$$

- A recurrent state is positive-recurrent if

$$E(T_j) = \sum_{n=1}^{\infty} nP(T_{jj} = n) < \infty.$$

We aim to compute:

- $E(T_{-+})$: the expected number of steps needed to reach the bull state $(+)$ from the bear state $(-)$.
- $E(T_{0+})$: the expected number of steps needed to reach the bull state $(+)$ from the stable state (0).

Using the first-step analysis, we obtain the following system of equations expressed in matrix form:

$$\begin{pmatrix} E(T_{-+}) \\ E(T_{0+}) \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 0.230769 & 0.317308 \\ 0.190871 & 0.522822 \end{pmatrix} \begin{pmatrix} E(T_{-+}) \\ E(T_{0+}) \end{pmatrix}.$$

Solving this we obtain,

$$\begin{pmatrix} E(T_{-+}) \\ E(T_{0+}) \end{pmatrix} = \begin{pmatrix} 1.546 & 1.028 \\ 0.618 & 2.491 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2.574 \\ 3.109 \end{pmatrix}.$$

The expected number of steps to transition from the bear state $(-)$ to the bull state $(+)$ is:

$$E(T_{-+}) = 2.574 \text{ weeks.}$$

The expected number of steps to transition from the stable state (0) to the bull state $(+)$ is:

$$E(T_{0+}) = 3.109 \text{ weeks.}$$

On average, it takes 2.574 weeks for the market to transition from a bearish state to a bullish state and 3.109 weeks for the market to transition from a stable state to a bullish state.

4.4 Mean Recurrence Time

The mean recurrence time for a state i in a Markov chain is the expected number of steps required to return to state i. It is given by:

$$E_i = \frac{1}{\pi_i}$$

For the bull state $(+)$, the mean recurrence time is $E_+ = \frac{1}{0.3365} \approx 2.97$ weeks. This implies that, in the long run, it takes approximately 3 weeks for the market to return to a bullish state after being in a bullish state.

5 Results

Our analysis showed that the market spent the majority of its time in the stable state, with smaller proportions of time in the bull and bear states. The stationary distribution, representing the long-term probabilities, indicated that the market is most likely to remain stable approximately 46% of the time, while the probabilities of being in a bull or bear state were 34% and 20%, respectively. To evaluate the effectiveness of our Markov model in predicting market behavior, we compared its stationary distribution to real historical market data from January 2023 to January 2025. The historical data was obtained from Yahoo Finance, categorizing each week as bullish, bearish, or stable. Over this period of 104 weeks, we observed the following distribution:

- 41 weeks were bullish (39%)
- 20 weeks were bearish (20%)
- 43 weeks were stable (41%)

These empirical results align closely with our predicted stationary distribution as shown in the Table 1.

	Prediction	Test Data
Bear	0.2	0.2
Stable	0.46	0.41
Bull	0.34	0.39

Table 1: Comparison of the derived long-term behavior and the test data.

6 Conclusion

In this report, we analyzed the S&P 500 index from January 5, 2013, to December 30, 2022, to model the behavior of the market using a Markov chain. By categorizing the market into three states—bull, bear, and stable—we were able to compute the transition matrix, revealing the probabilities of moving between these states.

Our findings showed that the market spent most of its time in the stable state, with a relatively smaller proportion of time in the bull and an even smaller proportion of time in the bear states. The stationary distribution, which gives the long-run probabilities of the market being in each state, indicated that the market is most likely to remain stable for around 46% of the time, which was somewhat accurate while evaluating with the last two years data.

However, the model has limitations. It assumes stationary transition probabilities, which may not hold in real-world financial markets that are often non-stationary. Additionally, the simplification of market conditions into three discrete states (bull, bear, and stable) does not fully capture the nuanced and continuous nature of market behavior.

Despite these limitations, the model provides a useful framework for understanding market behavior. Future work could explore alternative thresholds for categorizing market states or incorporate more advanced techniques like Hidden markov process(HMM) to better reflect the dynamics of smaller or more volatile markets. Overall, this study demonstrates the potential of Markov chains in modeling financial markets while highlighting areas for further refinement.

7 References

1. Lecture notes from Örjan Stenflo

2. Introduction to Probability Models by Sheldon M Ross

3. Introduction to Stochastic Process by Gregory R. Lawler

4. R. G. Gallager, "Finite-state Markov Chains" in Stochastic Processes: Theory For Applications. United Kingdom: Cambridge University Press, 2013, 160-213

Appendix

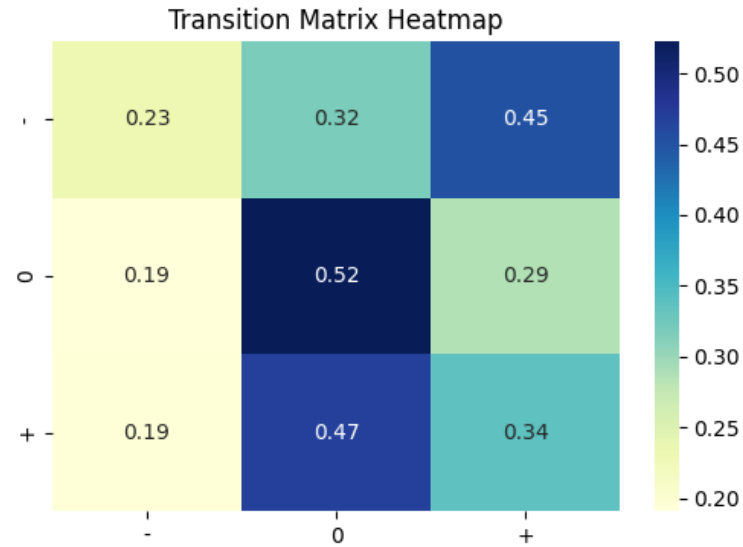

Figure 2: Transition matrix heatmap.

In this appendix, we have Python code used to perform data analysis. and plot Figures for the project.

```
import numpy as np
import pandas as pd
import yfinance as yf
import matplotlib.pyplot as plt
import seaborn as sns



# Fetch S&P 500 data for the last 10 years
ticker = "^GSPC"          # S&P 500 index
start_date = "2013-01-05"
end_date = "2022-12-30"


# Fetch the data using yfinance
data = yf.download(ticker, start=start_date, end=end_date, interval='1wk')


# Calculate weekly returns
# AS (close price this week - close price last week) / close price last week * 100
data['Return'] = data['Close'].pct_change() * 100



```

```python

# Drop the first and last row with NaN (as it has no return)
returns = data['Return'].dropna()



# Ensure 'returns' is a pandas Series (if it's in a DataFrame)
returns = returns.squeeze()  # Convert to Series if it's a single-column DataFrame





# Define market states based on returns
def classify_state(change):
    if change > 1:         # Bull market: price increase > 1%
        return "+"
    elif change < -1:       # Bear market: price decrease > 1%
        return "-"
    else:                   # Stable market: price change between -1% and 1%
        return "0"



# Apply classify_state function to each return value in 'returns' series
states = [classify_state(x) for x in returns]





# Define state space
state_space = ["-", "0", "+"]



# Initialize transition counts
transition_counts = {i: {j: 0 for j in state_space} for i in state_space}



# Count transitions
for i in range(len(states) - 1):
    current_state = states[i]     # Current state (it will be a string: "+", "0", or "-")
    next_state = states[i + 1]    # Next state (next return's classification)
    if current_state in state_space and next_state in state_space:
        transition_counts[current_state][next_state] += 1


# Compute transition probabilities
transition_matrix = np.zeros((3, 3))


# calculates the transition matrix by
# P_i =count the total weeks spend in one state
```

```python
# P_ij= count the total weeks from one state to another or to same state
# divide the  P_ij/P_i

for i, state_i in enumerate(state_space):
    total_transitions = sum(transition_counts[state_i].values())
    if total_transitions > 0:
        for j, state_j in enumerate(state_space):
            transition_matrix[i, j] = transition_counts[state_i][state_j] / total_transitions


# Convert to DataFrame for readability
transition_matrix_df = pd.DataFrame(transition_matrix, index=state_space, columns=state_space)



# Display transition matrix
print("\nTransition Matrix:")
print(transition_matrix_df)

# Plot the transition matrix as a heatmap
plt.figure(figsize=(6, 4))
sns.heatmap(transition_matrix_df, annot=True, cmap="YlGnBu", fmt=".2f")
plt.title("Transition Matrix Heatmap")

plt.show()

# Plot weekly returns over time
plt.figure(figsize=(10, 6))
plt.plot(data.index[1:], returns, label="Weekly Returns (%)", color="blue")
plt.axhline(y=1.0, color="green", linestyle="--", label="Bull Threshold (+1%)")
plt.axhline(y=-1.0, color="red", linestyle="--", label="Bear Threshold (-1%)")
plt.axhline(y=0.0, color="black", linestyle="-", label="Stable (0%)")
plt.title("S&P 500 Weekly Returns (jan 2013- dec 2022)")
plt.xlabel("Date")
plt.ylabel("Return (%)")
plt.legend()
plt.show()




# Count the number of weeks in each state (Bull, Bear, Steady)
bull_weeks = states.count("+")
bear_weeks = states.count("-")
steady_weeks = states.count("0")

# Total number of weeks (length of the 'returns' series)
total_weeks = len(returns)

# Print the results
print(f"Total number of weeks in the data: {total_weeks}")
print(f"Number of Bull market weeks: {bull_weeks}")
print(f"Number of Bear market weeks: {bear_weeks}")
print(f"Number of Steady market weeks: {steady_weeks}")
```

```python
from google.colab import files
files.download('trasnsitionmatrixheatmap.png')
files.download('weeklyreturns.png')
```

Listing 1: Python Code for data analysis and plotting figures